# MINHUI LI

Current 26 March 2026

School of the Environment
Yale University
21 Sachem Street ESC 338, New Haven, CT 06511
**Email**: minhui.li@yale.edu or minhui0603@gmail.com
**Web**: https://mhl-2024.github.io/

## Current Appointment

**School of the Environment, Yale University** — United States

Postdoctoral associate — *Advisor:* Peter Raymond — *Jan. 2025-today*

## Education

**Department of Hydraulic Engineering, Tsinghua University** — China

*Doctor of Engineering* — *Supervisors:* Xudong Fu & Baosheng Wu — *Aug. 2018-Jun. 2024*

**Dissertation Topic:** "Controls on Stream Networks and Drainage Basin Shapes"

**College of Harbor, Coastal and Offshore Engineering, Hohai University** — China

*Bachelor of Engineering* — *Sep. 2014-Jun. 2018*

## Professional Experiences

**Department of Environmental Systems Science, ETH Zurich** — Switzerland

| | | |
|---|---|---|
| Postdoctoral researcher | *Advisor:* James Kirchner | *Aug. 2024-Dec. 2024* |
| Scientific assistant | *Advisor:* James Kirchner | *Dec. 2023-Jul. 2024* |
| Visiting Ph.D. student | *Advisor:* James Kirchner | *Dec. 2021-Nov. 2023* |

## Research Interests

**Geomorphology, Hydrology and Biogeochemistry**

- Hydrological controls on stream network expansion and contraction and their implications for carbon dynamics and their implications for riverine carbon dynamics
- Continental- to Global-Scale morphological analysis of stream networks and watersheds
- Dynamic connectivity between landscapes and stream networks and their controls on hydrologic response across different climates

## Publications

Peer-reviewed journal papers

9. **Li, M.**, Seybold, H., Fu, X., Wu, B., Raymond, P., & Kirchner, J.W. (2026). Climate's influence on topography encoded in stream network topology and geometry. *Nature Communications.* (*Accepted*)

8. **Li, M.**, Seybold, H., Beria, H., Floriancic, M.,Fu, X., Raymond, P., & Kirchner, J.W. (2026). Controls on the magnitude and timing of runoff response to rainfall across the continental US. *Environmental*

*Research Letters, 21*, 054017.

7. **Li, M.**, Seybold, H., Wu, B., Chen, Y., Fu, X., & Kirchner, J.W. (2024). Global analysis of topographic and climatic controls on drainage basin shapes. *Geophysical Research Letters, 51*(8), e2023GL105804.

6. **Li, M.**, Seybold, H., Wu, B., Chen, Y., & Kirchner, J.W. (2023). Interaction between tectonics and climate encoded in the planform geometry of stream networks on the eastern Tibetan Plateau. *Geophysical Research Letters, 50*(14), e2023GL104121.

5. **Li, M.**, Wu, B., Chen, Y., & Li, D. (2022). Quantification of river network types based on hierarchical structures. *Catena*, *211*, 105986.

4. Ciais, P., Zhu, Y., Cai, Y., Lan, X., Michel, S., Zheng, B., Zhao, Y., Hauglustaine, D.A., Lin, X., Zhang, Y., Sun, S., Tian, X., Zhao, M., Wang, Y., Chang, J., Dou, X., Liu, Z., Andrew, R., Quinn, C.A., Poulter, B., Ouyang, Z., Yuan, W., Yuan, K., Zhu, Q., Li, F., Pan, N., Tian, H., Yu, X., Rocher-Ros, G., Johnson, M.S., Li, M., **Li, M.**, Feng, D., Raymond, P., Canadell, P., Jackson, R.B., Saunois, M., Bousquet, P., Peng, S. (2025). Why methane surged in the atmosphere during the early 2020s. *Science*, *391*(6785), eadx8262.

3. Zhang, B., Wu, B., Zhang, R., Ren, S., & **Li, M.** (2021). 3D numerical modelling of asynchronous propagation characteristics of flood and sediment peaks in Three Gorges Reservoir. *Journal of Hydrology*, *593*, 125896.

2. Chen, Y., Wu, B., Xiong, Z., Zan, J., Zhang, B., Zhang, R., Xue, Y., **Li, M.**, & Li, B. (2021). Evolution of eastern Tibetan river systems is driven by the indentation of India. *Communications Earth & Environment*, *2*(1), 256.

1. Qin, C., Wu, B., Wang, Y., Fu, X., Xue, Y., Li, D., **Li, M.**, & Zhang, Y. (2020). Dynamic variability of at-a-station hydraulic-geometry for mountain rivers in the southeast Qinghai-Tibet Plateau: The cases of Yalong River and upper Jinsha River. *Catena*, *194*, 104723.

Peer-reviewed journal papers in Chinese

3. **Li, M.**, Wu, B, & Chen Y (2022). Planform geometry and controlling factors of river networks in the Yellow River source zone. *Acta Geographica Sinica*, 77(11): 2878-2889.

2. **Li, M.**, Chen Y., & Wu, B**.** (2020). Analysis of features and factors controlling typical drainage networks in the Tibetan Plateau. *Journal of Tsinghua University* (*Science and Technology*), *60*(11), 951-957.

1. Chen, Y, Wu, B., & **Li, M.** (2022). Chemical weathering to climatic variations in the Yellow River source region. *Journal of Tsinghua University* (*Science and Technology*), *62*(12), 1945-1952.

## Presentations

| | |
|---|---|
| Dec. 2025 | **Li, M.**, Beria, H., Seybold, H., van Meerveld, I., Kirchner, J.W., & Raymond, P. Effects of Hillslope-network connectivity on rainfall runoff response. *American Geophysical Union Fall Meeting*, New Orleans, United States (poster) |
| Jun. 2025 | Li, M., Seybold, H., Beria, H., Floriancic, M.,Fu, X., Raymond, P., & Kirchner, J.W. Controls on the magnitude and timing of runoff response to rainfall across the continental US. *Gordon Research Conference on Catchment Science*, Andover, New Hampshire, United States (poster) |

| | |
|---|---|
| May 2024 (invited) | **Li, M.**, Controls on stream network and drainage basin shapes, Hohai University, Nanjing, China (talk) |
| Apr. 2024 | **Li, M.**, Seybold, H., Fu, X., Wu, B., & Kirchner, J.W. Climatic controls on stream network topology. *European Geosciences Union General Assembly*, Vienna, Austria (talk) |
| Dec. 2023 | **Li, M.**, Seybold, H., Wu, B., Chen, Y., Fu, X., & Kirchner, J.W. Topographic and climatic controls on global patterns in drainage basin shape. *American Geophysical Union Fall Meeting*, San Francisco, United States (talk) |
| Oct. 2023 (invited) | **Li, M.**, Controls on stream network and drainage basin shapes, Department of Earth and Planetary Sciences, ETH, Zurich, Switzerland (talk) |
| Apr. 2023 | **Li, M.**, Seybold, H., Wu, B., Chen, Y., & Kirchner, J.W. Interaction between tectonics and climate encoded in the planform geometry on the Tibetan Plateau. *European Geosciences Union General Assembly*, Vienna, Austria (talk) |
| May 2022 | **Li, M.**, & Wu, B. Planform geometries and controlling factors of river networks in the Yellow River source zone. *European Geosciences Union General Assembly*, Vienna, Austria (talk) |
| May 2020 | **Li, M.**, Wu, B., & Chen, Y. Features and controlling factors of drainage networks in the Tibetan Plateau. *European Geosciences Union General Assembly*, virtual (talk) |
| May 2026 | Brinkerhoff, C., Raymond, P., Flores, J., **Li, M.**, Zhang, S., Sun, X., & Feng, D. Using SWOT to revisit global river gas exchange. *European Geosciences Union General Assembly*, Vienna, Austria |
| Dec. 2025 | Allen, G., **Li, M.**, & Go, D. Variability of stream and river inundation across scales. *American Geophysical Union Fall Meeting*, New Orleans, United States (poster) |
| Dec. 2025 | Brinkerhoff, C., Raymond, P., Shrestha, B., & **Li, M.** Informing non-perennial stream models with in-situ sensing. *American Geophysical Union Fall Meeting*, New Orleans, United States (poster) |
| Dec. 2023 | Seybold, H., **Li, M.**, & Kirchner, J.W. Runoff controls on stream network branching. *American Geophysical Union Fall Meeting*, San Francisco, United States (talk) |

## Mentoring

| | |
|---|---|
| 2026 | *Undergraduate* Bryan Bautista (Central Connecticut State University) |
| 2026 | *Undergraduate* Yazim Alicea Roman (Southern Connecticut State University) |

## Academic Services

### Journal Reviewer

Water resources research; Journal of geophysical research: Earth Surface; Geomorphology

## Skills

### Program

- Proficient in geospatial data analysis using Python and ArcGIS
- Experienced in working with Linux operating system and HPC

- Expertise in working with MATLAB and R

**Language**

- Chinese: Native.
- English: Fluent. Certificate: Level 5 of the Public English Test System (PETS5)

## Honors and Awards

| | |
|---|---|
| Dec. 2022 | The Second Prize Scholarship (Tsinghua University) |
| Apr. 2021 | Best Paper Award (Tsinghua University) |
| Dec. 2019 | The Second Prize Scholarship (Tsinghua University) |
| Jun. 2018 | Outstanding Graduate (Hohai University) |
| Apr. 2017 | Meritorious Winner in Interdisciplinary Contest In Modeling |
| Aug. 2016 | First Prize in the 13th Mathematics Competition |
| Jul. 2016 | Second Prize of National Mathematic Modeling Competition |
| Nov. 2015 | Academic Excellence Scholarship (Hohai University) |